

Index of key relations & assumptions

For an ideal, calorifically perfect gas.

$$\gamma = C_p / C_v \quad [NA]$$

$$C_p - C_v = R \quad [J/kg \cdot K]$$

$$\begin{aligned} C_p &= \frac{\gamma R}{\gamma - 1} \\ C_v &= \frac{R}{\gamma - 1} \end{aligned}$$

specific volume:

$$v = \frac{V}{m} = \frac{1}{\rho} \quad [m^3/kg]$$

Generally

$$a^2 = \left(\frac{\partial p}{\partial \rho} \right)_s$$

For an ideal gas

$$a = \sqrt{\gamma \frac{p}{\rho}} = \sqrt{\gamma R T} \quad [m/s]$$

Specific total Energy & Enthalpy

$$E = e + \frac{1}{2} |\vec{u}|^2 \quad [J/kg]$$

$$H = h + \frac{1}{2} |\vec{u}|^2 = E + p v \quad [J/kg]$$

ΔS for a reversible, ideal, C.P. gas: $\Delta S = S - S_0 = C_v [\ln(p v^\gamma) - \ln(p_0 v_0^\gamma)] \quad [J/kg \cdot K]$

Further, if flow is isentropic:

$$\frac{T_2}{T_1} = \left(\frac{p_2}{p_1} \right)^{\frac{\gamma-1}{\gamma}} = \left(\frac{\rho_2}{\rho_1} \right)^{\gamma-1} = \left(\frac{v_2}{v_1} \right)^{-(\gamma-1)} \quad \star \text{ isentropic relation!}$$

For ideal, C.P. gas: $p = (\gamma - 1) p e$

Euler equations ($Kn \rightarrow 0$)

$$\frac{\partial}{\partial t} \begin{bmatrix} \rho \\ \rho \vec{u} \\ \rho E \end{bmatrix} + \nabla \cdot \begin{bmatrix} \rho \vec{u} \\ \rho \vec{u} \otimes \vec{u} + p \vec{I} \\ \rho \vec{u} H \end{bmatrix} = 0$$

NS equations ($Kn \in [0.001, 0.1]$)

$$\frac{\partial}{\partial t} \begin{bmatrix} \rho \\ \rho \vec{u} \\ \rho E \end{bmatrix} + \nabla \cdot \begin{bmatrix} \rho \vec{u} \\ \rho \vec{u} \otimes \vec{u} + p \vec{I} - \vec{\tau} \\ \rho \vec{u} H + \vec{q} \end{bmatrix} = 0$$

Key difference terms to Euler eqns.

Material derivative:

$$\frac{D\phi}{Dt} = \frac{\partial \phi}{\partial t} + \vec{u} \cdot \nabla \phi$$

local, Eulerian ROC.

convective ROC.

A Lagrangian formulation where you follow ^{particle} ϕ instead of a control volume (x, y, z) across time.

A review of UG fluids

$\nabla \cdot \vec{u} = 0$ incompressibility

$\nabla \times \vec{u} = 0$ irrotationality $\rightarrow$ implies $\vec{u} = \nabla \phi$ since $\nabla \times (\nabla \phi) = 0$ is a mathematical property (so is $\nabla \cdot (\nabla \times \phi) = 0$)

$\Rightarrow \nabla \cdot (\nabla \phi) = 0 \Rightarrow \nabla^2 \phi = 0 \rightarrow$ implies irrot. & incomp.

$\phi \triangleq$ velocity potential

For incompressible flows, can define a stream function $\psi(x, y)$ $\frac{dy}{dx} = \frac{v}{u}$

$$u_1 = \frac{\partial \psi}{\partial y}$$

$$u_2 = -\frac{\partial \psi}{\partial x}$$

$\rightarrow$ preserves $\nabla \cdot \vec{u} = \frac{\partial u_1}{\partial x} + \frac{\partial u_2}{\partial y} = 0$ when ψ is const

STAGNATION PROPERTIES

@ stagnation, $u=0$, $H_0 = C_p T_0$

$$\frac{T_0}{T} = 1 + \frac{u^2}{2C_p T} = 1 + \frac{\gamma-1}{2} M^2$$

(2nd half ideal assumption)

$$\frac{p_0}{p} = \left(1 + \frac{\gamma-1}{2} M^2\right)^{\frac{\gamma}{\gamma-1}}$$

$$\frac{\rho_0}{\rho} = \left(1 + \frac{\gamma-1}{2} M^2\right)^{\frac{1}{\gamma-1}}$$

normal shock conservation equations

$$\rho_1 u_1 = \rho_2 u_2$$

$$\rho_1 u_1^2 + p_1 = \rho_2 u_2^2 + p_2$$

$$\rho_1 u_1 H_1 = \rho_2 u_2 H_2$$

Aside on γ

$$\gamma = \frac{N_{\text{DOF}} + 2}{N_{\text{DOF}}} = \frac{C_p}{C_v}$$

$$= \frac{5}{3} \text{ (monoatomic)} \quad \text{or} \quad \frac{7}{5} \text{ (diatomic)}$$

Rankine Hugoniot Relations for Stationary & Moving normal shocks.

$$\frac{p_2}{p_1} = 1 + \gamma M_1^2 \left(1 - \frac{u_2}{u_1}\right) = \frac{2\gamma M_1^2 - (\gamma-1)}{\gamma+1}$$

$$\frac{\rho_2}{\rho_1} = \frac{u_1}{u_2} = \frac{(\gamma+1)M_1^2}{2 + (\gamma-1)M_1^2}$$

$$\frac{T_2}{T_1} = \frac{p_2}{p_1} \frac{\rho_1}{\rho_2} = \left[1 + \frac{2\gamma}{\gamma+1} (M_1^2 - 1)\right] \left[\frac{2 + (\gamma-1)M_1^2}{(\gamma+1)M_1^2}\right]$$

strong shock? $M_1 \rightarrow \infty$

$$\frac{p_2}{p_1} \rightarrow \infty, \quad \frac{T_2}{T_1} \rightarrow \infty, \quad \frac{\rho_2}{\rho_1} \rightarrow \frac{\gamma+1}{\gamma-1}$$

* finite density property!

$$M_2 \rightarrow \sqrt{\frac{\gamma-1}{2\gamma}}$$

$$\frac{\tilde{u}_1}{\tilde{u}_2} \rightarrow \frac{\gamma+1}{\gamma-1}$$

$$M_2^2 = \frac{2 + (\gamma-1)M_1^2}{2\gamma M_1^2 - (\gamma-1)}$$

* the most important

weak shock? $\frac{p_2 - p_1}{p_1} \ll 1$

$$\frac{p_2 - p_1}{p_1} \ll 1$$

Taylor expansion??

$$S_2 - S_1 = C_p \ln\left(\frac{T_2}{T_1}\right) - R \ln\left(\frac{p_2}{p_1}\right)$$

* for all relations here, use $\tilde{u} = u - c_s$ and $\tilde{M} = \tilde{u}/a$ for moving shocks, same relations hold!

Riemann Problem & M- Σ problems

$$M \triangleq \frac{\Delta u}{a_0}$$

$$\Sigma \triangleq \frac{\Delta p}{\rho_0}$$

relations in terms of M, Σ

$$\text{cons. of mass: } \frac{\partial \Sigma}{\partial t} + a_0 \frac{\partial M}{\partial x} = 0$$

$$\text{cons. of mom.: } \frac{\partial M}{\partial t} + a_0 \frac{\partial \Sigma}{\partial x} = 0$$

$$\textcircled{1} + \textcircled{2}: \frac{\partial}{\partial t} (M + \Sigma) + a_0 \frac{\partial}{\partial x} (M + \Sigma) = 0$$

$$\textcircled{2} - \textcircled{1}: \frac{\partial}{\partial t} (M - \Sigma) - a_0 \frac{\partial}{\partial x} (M - \Sigma) = 0$$

* gives rise to P^+ , P^- curves with slope a_0

$$\frac{\partial J^\pm}{\partial t} \pm a_0 \frac{\partial J^\pm}{\partial x} = 0$$

rewritten

Riemann invariants!

$$J^\pm = M \pm \Sigma$$

Geometric middle-state soln. using $P^\pm$ on $x-t$ diag.

$$M_P = \frac{M_A + \Sigma_A}{2} + \frac{M_B - \Sigma_B}{2}; \quad \Sigma_P = \frac{M_A + \Sigma_A}{2} - \frac{M_B - \Sigma_B}{2}$$

$$x_P = \frac{x_A + x_B}{2}; \quad t_P = \frac{x_B - x_A}{2a_0}$$

Rayleigh Flow

$$\frac{p_2}{p_1} = \frac{1 + \gamma M_1^2}{1 + \gamma M_2^2} = \frac{p_{2T_2}}{p_{1T_1}} = \frac{u_1 T_2}{u_2 T_1}$$

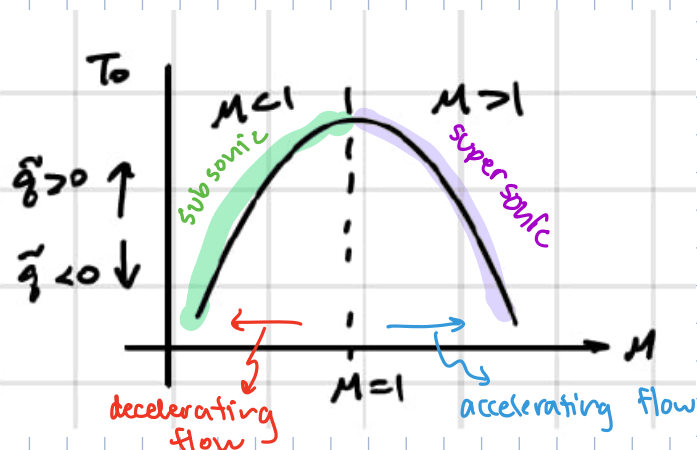
$$\frac{u_1}{u_2} = \frac{M_1 a_1}{M_2 a_2} = \frac{M_1}{M_2} \sqrt{\frac{T_1}{T_2}}$$

$$\frac{T_2}{T_1} = \left(\frac{p_2}{p_1}\right)^2 \left(\frac{M_2}{M_1}\right)^2$$

$$\frac{T_{0,2}(M_2)}{T_{0,1}(M_1)} = \frac{T_2}{T_1} \left(\frac{1 + \frac{\gamma-1}{2} M_2^2}{1 + \frac{\gamma-1}{2} M_1^2} \right)$$

$$\frac{T_0(M)}{T_0^*} = \left[2 + (\gamma-1)M^2 \right] \frac{(1+\gamma)M^2}{(1+\gamma M^2)^2}$$

$$\tilde{q} = C_p [T_{0,2} - T_{0,1}]$$

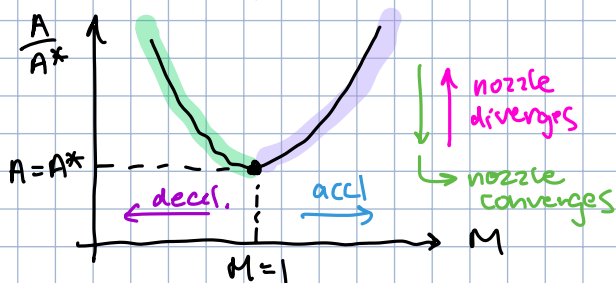


Mach-velocity: $\frac{du}{u} = (M^2 - 1)^{-1} \frac{dA}{A}$

Mach-area:

$$\frac{A}{A^*} = \frac{\left(1 + \frac{\gamma-1}{2} M^2\right)^{\frac{\gamma+1}{2(\gamma-1)}}}{\left(\frac{\gamma+1}{2}\right)^{\frac{\gamma+1}{2(\gamma-1)}}} \frac{1}{M}$$

* from this, we can infer accel, decel of a converging-diverging nozzle.



Mass flow:
(in normal
& throat
conditions)

$$\dot{m} = \frac{P}{RT} (M \sqrt{\gamma R T}) = \sqrt{\frac{\gamma}{R}} \frac{P}{\sqrt{T}} M A = \sqrt{\frac{\gamma}{R}} \frac{P}{\sqrt{T}} M A^* \frac{A}{A^*}$$

$$\dot{m} = \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{2(\gamma-1)}} \sqrt{\frac{\gamma}{R}} \frac{P_0}{\sqrt{T_0}} A^*$$

Fanno Flow (with skin friction)

setup $(p_2 u_2^2 - p_1 u_1^2) A = -(p_1 - p_2) A - \int_0^L \pi D T_w(x) dx$

where $T_w \triangleq C_F \frac{1}{2} \rho u^2 \Rightarrow C_F = \frac{T_w}{\frac{1}{2} \rho u^2}$ [NA]

is the skin friction coefficient

Key Fanno Relation (& analytical integral)

$$\Phi_2 - \Phi_1 = \int_{x_1}^{x_2} \frac{4C_F}{D} dx = \int_{M_1}^{M_2} \frac{2(1-M^2)}{\gamma M^2} \left[1 + \frac{(\gamma-1)}{2} M^2 \right]^{-1} \frac{dM}{M}$$

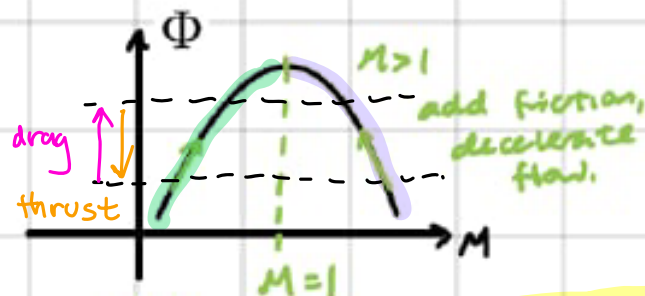
$$= \left[-\frac{1}{\gamma M^2} - \frac{\gamma+1}{2\gamma} \ln \left(\frac{M^2}{1 + \frac{\gamma-1}{2} M^2} \right) \right]_{M_1}^{M_2}$$

Fanno Flow across $M^*=1$ choke disc.

$$\int_0^L \frac{4C_F}{D} dx = \left(\int_0^{L^*} - \int_L^{L^*} \right) \frac{4C_F}{D} dx$$

$$\Rightarrow \frac{4\bar{C}_F L}{D} = \frac{4\bar{C}_F L^*(M_1)}{D} - \frac{4\bar{C}_F L^*(M_2)}{D}$$

* note sign conventions b/c of $\pm \pi$ definition



* subsonic behavior is counterintuitive, supersonic behavior is the inverse.

[Quasi-1D] converging nozzle:

$$Me = \sqrt{\frac{2}{\gamma-1} \left[\left(\frac{p_t}{p_{amb}} \right)^{\frac{\gamma-1}{\gamma}} - 1 \right]}$$

Mach @ exit

$$\frac{p_t}{p_{amb}} = \left(\frac{\gamma+1}{2} \right)^{\frac{\gamma}{\gamma-1}}$$

set $Me = 1$,
choke pressure ratio

This equation also works for the converging section of conv/div nozzles.

but use $\frac{p_t}{p^*}$ since p^* is choke pressure & p_{amb} is @ exit.

Nonlinear Problems

$$\frac{\partial J^\pm}{\partial t} + (u \pm a) \frac{\partial J^\pm}{\partial x} = 0 \rightarrow$$

characteristic velocities!

$$\frac{\partial s}{\partial t} + u \frac{\partial s}{\partial x} = 0$$

$$J^\pm = u \pm \frac{2a}{\gamma-1} = \text{const}$$

$$\Gamma^\pm = u \pm a$$

nonlinear Riemann invariants / characteristics

Hugoniot curves for left/right moving shocks.

$$C_s = \pm a_{\text{pre}} \sqrt{1 + \frac{\gamma+1}{2\gamma} \frac{p_{\text{post}} - p_{\text{pre}}}{p_{\text{pre}}}}$$

$$p_{\text{post}} - p_{\text{pre}} = \pm m \Delta u = \pm \rho_4 a_4 \sqrt{1 + \frac{\gamma+1}{2\gamma} \frac{p_{\text{post}} - p_{\text{pre}}}{p_{\text{pre}}}} (u_3 - u_4)$$

$$\Delta p \propto \Delta u$$

$$\Delta p \propto -\Delta u$$

under weak shock limit $\frac{p_{\text{post}} - p_{\text{pre}}}{p_{\text{pre}}} \ll 1$, it can be shown using limits & isentropic relation $\Delta p = a^2 \Delta \rho$ that Hugoniot curve becomes $M-\Sigma=0$!

Poisson curves for left/right moving expansion fans

$$u_{\text{post}} - u_{\text{pre}} = \pm \frac{2a_{\text{pre}}}{\gamma-1} \left[\left(\frac{p_{\text{post}}}{p_{\text{pre}}} \right)^{\frac{\gamma-1}{2\gamma}} - 1 \right]$$

pseudo Δp term

$\Delta p \propto \Delta u$; J^- is const across fan

$\Delta p \propto -\Delta u$; J^+ is const acr. fan

4 important Observations comparing P & H.

① as $p_{\text{post}} \rightarrow \infty$ $H: p_{\text{post}} \propto (u_{\text{post}} - u_{\text{pre}})^2$

$$P: p_{\text{post}} \propto (u_{\text{post}} - u_{\text{pre}})^{\frac{2\gamma}{\gamma-1}}$$

② as $p_{\text{post}} \rightarrow 0$
(vacuum cond.)

$$H: u_{\text{post min}} = u_{\text{pre}} - a_{\text{pre}} \sqrt{\frac{2}{\gamma(\gamma-1)}}$$

$$P: u_{\text{post min}} = u_{\text{pre}} - a_{\text{pre}} \frac{2}{\gamma-1}$$

③ at a reference point $u_{\text{pre}}, p_{\text{pre}}, \rho_{\text{pre}}, a_{\text{pre}}$, H & P share the same slope

$$\left. \frac{\partial p^*}{\partial u^*} \right|_{p^* \rightarrow p_{\text{pre}}, u^* \rightarrow u_{\text{pre}}} = \rho_{\text{pre}} a_{\text{pre}} \text{ for BOTH H \& P}$$

④ P has higher curvature than H on 3rd order effects
(derivation in notes).

Chapter 1 : Microscopic View of Fluids & Fluids Review

- see notes for VDFs & $\vec{u}$, ε (bulk velocity / mean translational)

mean free path: $\lambda_{mfp} = \frac{1}{8n\sigma}$
[m]

$\delta = 1$ for same-species collisions
 $\delta = \sqrt{2}$ for cross-species collisions
 $n \triangleq$ number density
 $\sigma \triangleq$ collisional cross-sectional area.

Knudsen #: $Kn = \frac{\lambda_{mfp}}{L}$ $L \triangleq$ characteristic length.
[NA]

Knudsen # intuitively measures whether something is collisional.

$Kn < 1 \rightarrow$ collisional

$Kn \sim 1 \rightarrow$ rarified gas

$Kn > 1 \rightarrow$ collisionless

the only condition where Euler / NS eqns are valid.

From normalized VDFs, and applying Boltzmann's equation, we get

EXACT FLUID EQNS.

cons of mass:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{u}) = \dot{S}$$

where $\dot{S}$ is any source/sink of mass.

cons of momentum:

$$\frac{\partial}{\partial t}(\rho \vec{u}) + \nabla \cdot (\rho \vec{u} \otimes \vec{u} + \bar{\mathbf{p}}) = \vec{f}$$

where $\vec{u} \otimes \vec{u}$ is the outer-product tensor of $\vec{u}$ & $\bar{\mathbf{p}}$ is the pressure tensor.

$\vec{f}$ is the body force

cons of energy:

$$\frac{\partial}{\partial t}(\rho E) + \nabla \cdot (\rho \vec{u} E + \bar{\mathbf{p}} \cdot \vec{u} + \vec{q}) = \dot{S}_e$$

where $\dot{S}_e$ is any source/sink of energy.

$\bar{\mathbf{p}}$ contains term $\bar{\mathbf{\tau}}$ & $\vec{q}$ make exact equations unclosed, hence we use Euler / NS.

Euler: $\bar{\mathbf{\tau}} \triangleq \bar{\mathbf{0}}$, $\vec{q} \triangleq \mathbf{0}$, $Kn \rightarrow 0$ (fully collisional)

Knudsen conditions.

NS: $\bar{\mathbf{\tau}} \triangleq \mu(\nabla \vec{u} + \nabla \vec{u}^T) + (\lambda_B - \frac{2}{3}\mu)(\nabla \cdot \vec{u})\bar{\mathbf{I}}$

closure expressions

$$\vec{q} \triangleq -K \nabla T$$

$Kn \in [0.001, 0.1]$ (finite collisionality)

* full expressions in key relations index.

Index Notation: fuck that shit... just know index pairs are localized within a term.

remember me?

$$\frac{\partial}{\partial t}(\rho u_i) + \nabla \cdot (\rho u_i \otimes \vec{u} + p \bar{\mathbf{I}}) = 0 \Rightarrow \frac{\partial}{\partial t}(\rho u_i) + \frac{\partial}{\partial x_j}(\rho u_i u_j + p \delta_{ij})$$

Chapter 2: Review of Thermodynamics

1st Law: $dE_{\text{tot}} = \delta Q - \delta W$ where $\delta W \triangleq p dv$

using specific quantities

$$de = \delta q - p dv \quad \text{bang!}$$

Recall enthalpy - energy relation $h = e + pv$

$$\delta q = de + p dv = dh - v dp$$

* important for C_p C_v definitions.

if set $dv = 0 \rightarrow C_v = \left. \frac{\partial q}{\partial T} \right|_v = \left. \frac{\partial e}{\partial T} \right|_v$

set $dp = 0 \rightarrow C_p = \left. \frac{\partial q}{\partial T} \right|_p = \left. \frac{\partial h}{\partial T} \right|_p = \left. \frac{\partial e}{\partial T} \right|_p + \left. \frac{\partial pv}{\partial T} \right|_p$

because highlighted term is positive for gases, $C_p \geq C_v$
is 0 for fluids (incomp) $C_p = C_v$

2nd Law:

Reversible process: $\frac{\delta Q}{T} = ds$; $\frac{\delta q}{T} = ds$; $ds = \frac{de}{T} + \frac{p}{T} dv = \frac{dh}{T} - \frac{v}{T} dp$

Adiabatic process: $\delta Q = 0$

$\Downarrow$

isentropic process: $ds = 0$; $pv^\gamma = p\rho^{-\gamma} = \text{const}$

* see relations index for important results.

Chapter 3: Fluid eqns, Discontinuities, & x-t diagrams.

Integral form of fluid eqns $\rightarrow$ we never use this.

Derivative form of " " $\rightarrow$ 2 different flavors

in conservative variables $(\rho, \rho \vec{u}, \rho E)$

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{u}) = 0$$

$$\frac{\partial}{\partial t}(\rho \vec{u}) + \nabla \cdot (\rho \vec{u} \otimes \vec{u} + p \vec{I}) = 0 \quad \Rightarrow$$

$$\frac{\partial}{\partial t}(\rho E) + \nabla \cdot (\rho \vec{u} H) = 0$$

in primitive variables $(\rho, \vec{u}, p)$
or $(\rho, \vec{u}, p)$

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \vec{u} = 0$$

$$\rho \frac{D\vec{u}}{Dt} + \nabla p = 0$$

$$\rho \frac{De}{Dt} + p \nabla \cdot \vec{u} = 0$$

* This $\Rightarrow$ conversion is in HW 1 & requires some mathematical gymnastics.

Rankine - Hugoniot relations in index:

NOTE ON ENTROPY CONDITION (*see $S_2 - S_1$ relation)

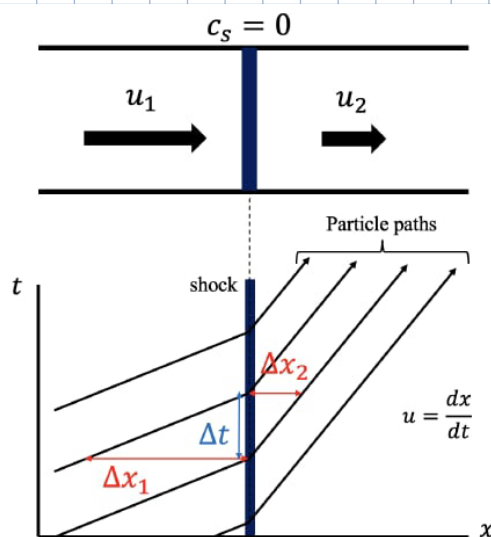
if $M_1 = 1$, $M_2 = 1$, $S_2 = S_1$ ✓ $\Rightarrow$ PERMISSIBLE! ($S_2 \geq S_1$ ok)

if $M_1 > 1$, $M_2 < 1$, $S_2 > S_1$ ✓

if $M_1 < 1$, $M_2 > 1$, $S_2 < S_1$ ✗ $\rightarrow$ not permissible, entropy cannot increase!

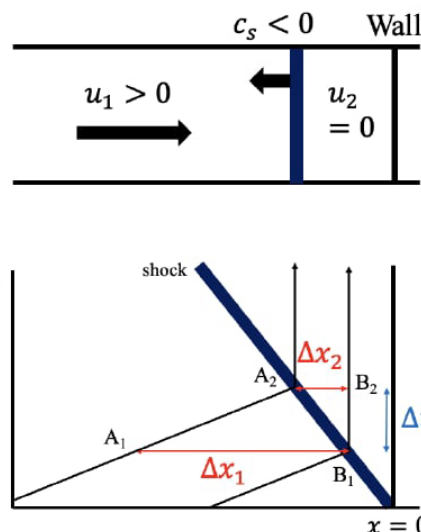
$$S_2 - S_1 = C_p \ln \left(\frac{T_{0,2}}{T_{0,1}} \right) - R \ln \left(\frac{p_{0,2}}{p_{0,1}} \right) \quad \text{where } \frac{T_{0,2}}{T_{0,1}} = 1 \text{ for adiabatic processes.}$$

x-t diagrams



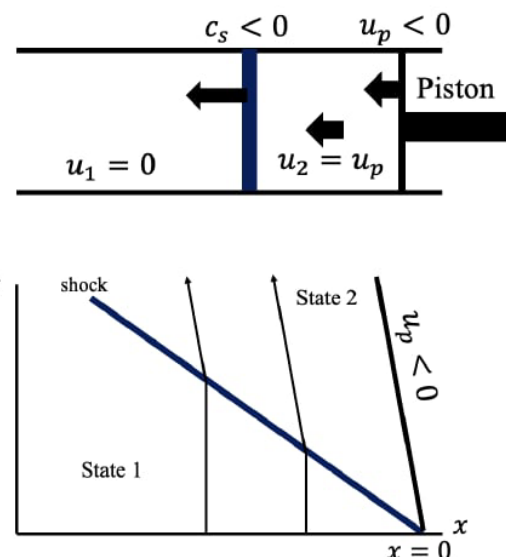
Stationary shock

$$\rho_1 \Delta x_1 = \rho_2 \Delta x_2$$



moving shock w/wall

$$c_s = \frac{\rho_1 u_1 - \rho_2 u_2}{\rho_1 - \rho_2} = \frac{\rho_1 u_1}{\rho_1 - \rho_2}$$



moving piston

Chapter 4: Riemann Problem, Method of characteristics, M-Σ problems

4.1: Waves, Phase, & Group velocities

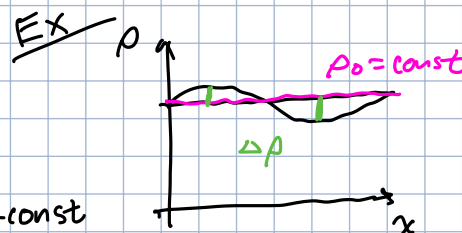
4.2: linear perturbations:

$$p = p_0 + \Delta p$$

$$\rho = \rho_0 + \Delta \rho$$

$$u = u_0 + \Delta u$$

+ Assume homentropy
 $s(x, y, z, t) = \text{const}$



homentropy $\iff a = a_0$ is constant!

* key M & I definitions arise $\rightarrow$ see key relations

Riemann Invariant Problems Tips

x-t diagram

- ① $\Gamma^\pm$ lines always have slope $\pm a_0$
- ② speed of shock itself also slope $\pm a_0$
- ③ middle state ②, ③ are the same for linear problems
- ④ measuring Δx distances across fixed time t and comparing them to Δx at t_0 gives density and hence compression/expansion intuition.
- ⑤ Path lines NEVER cross

M-Σ diagrams

- ① slope along invariants is ALWAYS ± 1 by invariant definition
- ② going from 1 $\rightarrow$ 2 & 4 $\rightarrow$ 3 is compression if $\Delta \Sigma > 0$ & expansion if $\Delta \Sigma < 0$

M-x or Σ-x diagrams

- ① cutoff points are at $a_0 t$ where t is honestly arbitrary. (tracks across $t = t_0$ on x-t diagram)

Chapter 8 : Viscous Flows, Entropy transport, Rayleigh, Fanno

* See lecture 7 notes for augmented NS with the following additional relations:

$$(1) \tau_{ij} = 2\mu S_{ij} + \lambda \delta_{ij} S_{kk} \quad \text{where } \mu \triangleq \text{1st viscosity} \\ \lambda \triangleq \text{2nd viscosity} \quad [N/m^2, Pa]$$

$$(2) S_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right), \quad S_{kk} = \frac{1}{2} \left(\frac{\partial u_k}{\partial x_k} + \frac{\partial u_k}{\partial x_k} \right) = \frac{\partial u_k}{\partial x_k} = \nabla \cdot \vec{u}$$

"rate of strain tensor" $[1/s]$

$$(3) K = u^2/2 \quad \text{"kinetic drift"}$$

$$(4) \Phi = \tau_{ij} \frac{\partial u_i}{\partial x_j} \geq 0$$

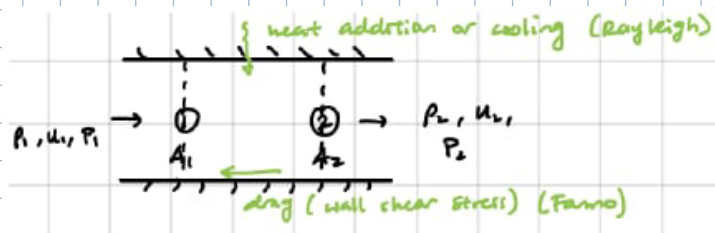
"viscous dissipation"

Entropy transport: what is the PDE that governs how specific entropy s moves under viscous ($\tilde{\tau} \neq 0$) & non-adiabatic ($\tilde{q}_i \neq 0$) conditions?

Final relation:

$$\frac{\partial}{\partial t}(\rho s) + \frac{\partial}{\partial x_j}(\rho u_j s + \frac{\tilde{q}_j}{T}) = \frac{K}{T^2} \left(\frac{\partial T}{\partial x_j} \right)^2 + \frac{\Phi}{T} \geq 0$$

General Form of 1-D Flow



cons eqns.

$$\rho_1 u_1 = \rho_2 u_2$$

$$p_1 + \rho_1 u_1^2 + \tilde{F} = p_2 + \rho_2 u_2^2$$

$$h_1 + \frac{1}{2} u_1^2 + \tilde{q} = h_2 + \frac{1}{2} u_2^2$$

when $\tilde{F} = 0$, $\tilde{q} = 0 \rightarrow$ Rankine-Hugoniot Jump eqns.

$\tilde{F} \neq 0$, $\tilde{q} = 0 \rightarrow$ Fanno flow

$\tilde{F} = 0$, $\tilde{q} \neq 0 \rightarrow$ Rayleigh flow

* See index for key Rayleigh & Fanno relations

Chapter 9 : Quasi-1D flow

* see notes for converging-diverging nozzle shock location cases.

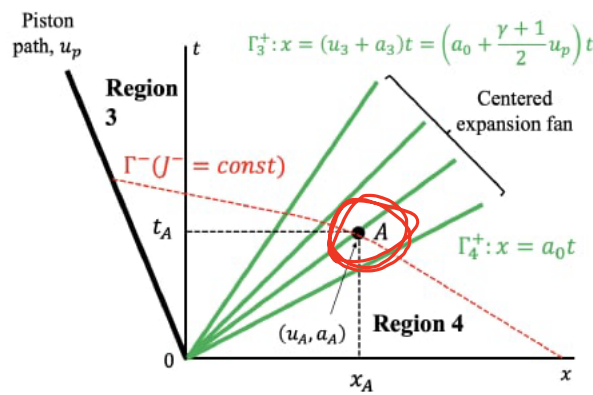
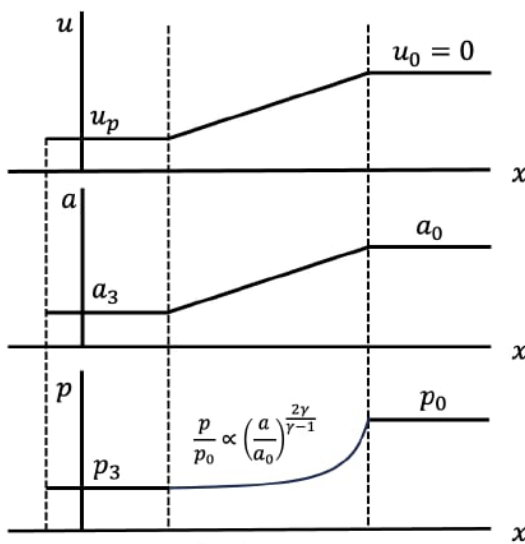
* see index for key relations

* pro tip: chamber pressures in nozzle problems can be treated as stagnation pressures so we can use stagnation properties.

Chapter 5 : 1D nonlinear, unsteady flow

Below: states WITHIN an isentropic expansion fan.

shocks are NOT isentropic!



$$u_A(x_A, t_A) = \frac{2}{\gamma+1} \left(\frac{x_A}{t_A} - a_0 \right)$$

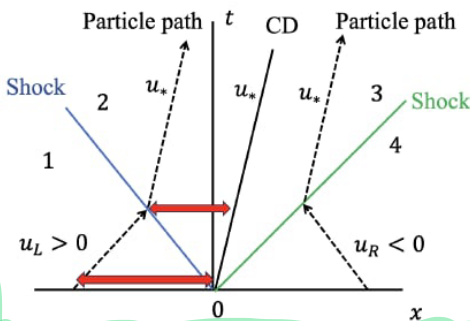
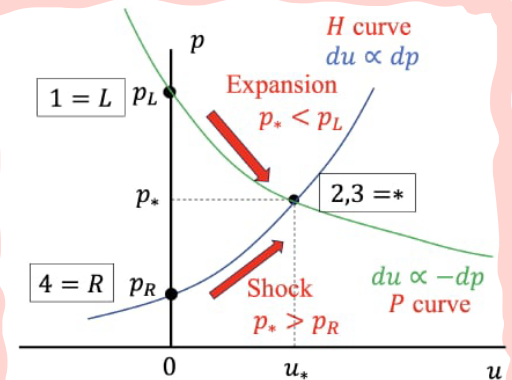
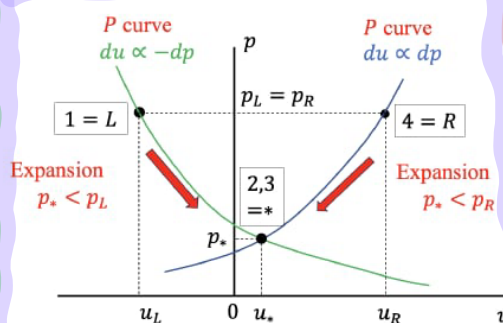
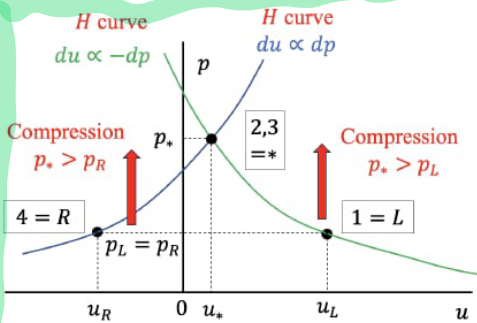
$$a_A(x_A, t_A) = \frac{\gamma-1}{\gamma+1} \frac{x_A}{t_A} + \frac{2}{\gamma+1} a_0$$

$$u_p t_A = \frac{0}{\gamma} \left(a_0 + \frac{\gamma+1}{2} u_p \right) t_A$$

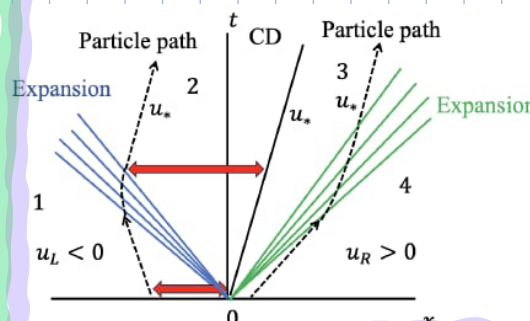
bounds of the expansion shock

- Tips:
- ① for Hugoniot curves, $\Gamma^\pm$ are discontinuous across shocks (non isentropic)
for Poisson curves, you can have $\Gamma^\pm_{pre} = \Gamma^\pm_{post}$!
 - ② u-p diagrams are effectively the M-Z diagrams of nonlinear problems.
 - ③ If $\Delta p < 0 \rightarrow$ curve is expansion fan $\rightarrow$ Poisson curve
If $\Delta p > 0 \rightarrow$ curve is shock $\rightarrow$ Hugoniot curve

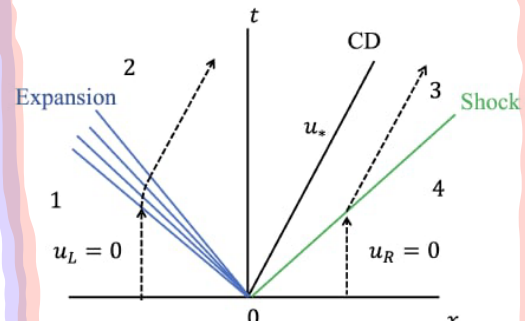
Examples



2 SHOCKS



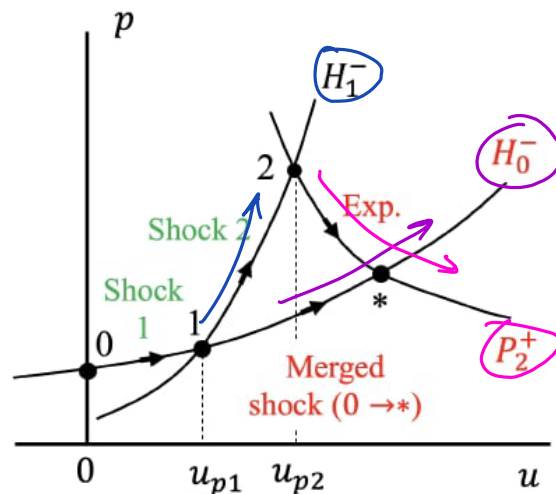
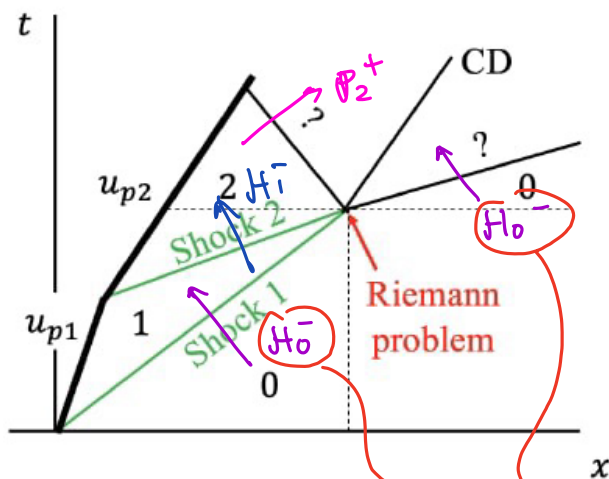
2 EXP. FANS



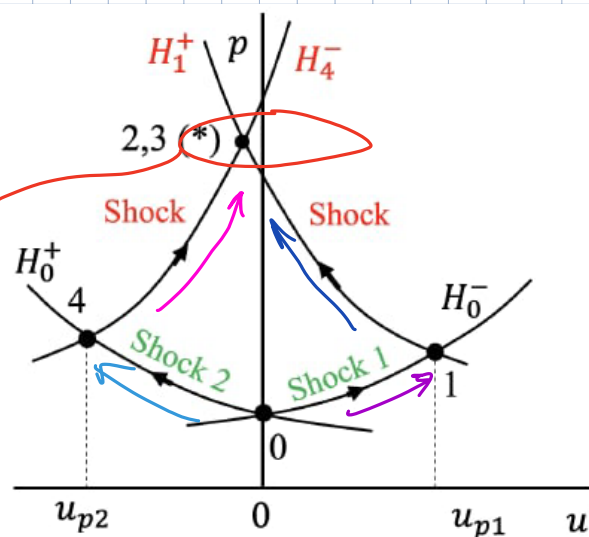
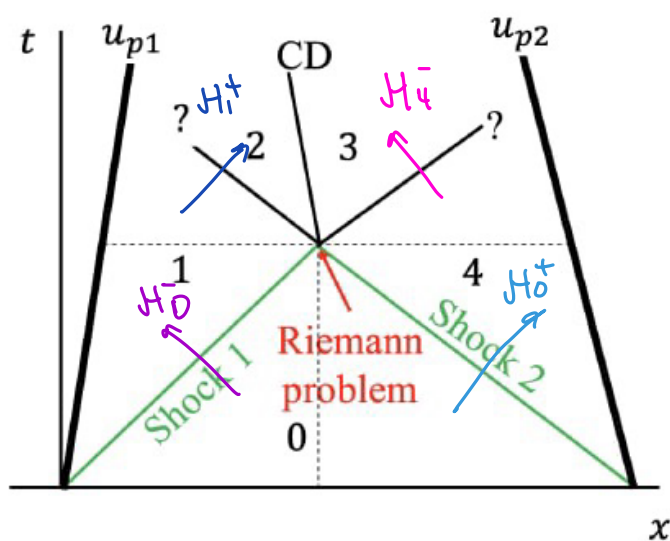
1 SHOCK
2 EXP

Chapter 6: 1-D nonlinear unsteady flow 2: The Riemann Problem

2 colliding Shocks (2 cases)



notice identical!

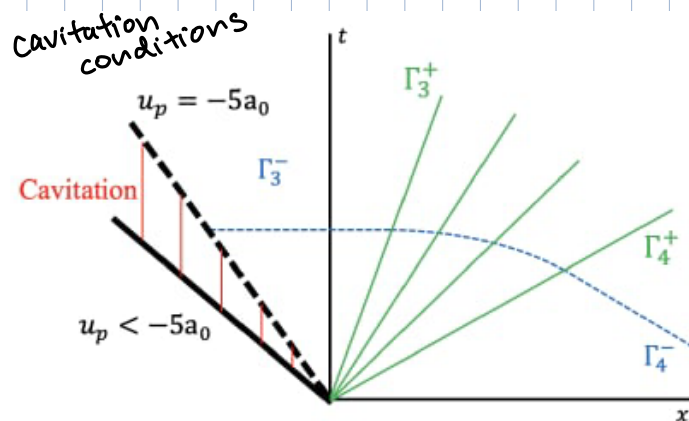
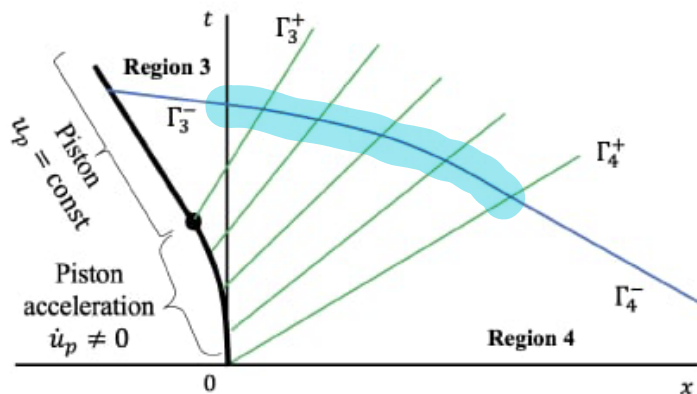


The actual signage of u^* is indeterminate.

PROBLEM TO EXPECT: expfan - expfan collision or exp-fan - shock collision.

Finite acceleration problems

Expansions



$$J^- = u - \frac{2a}{\gamma-1} = \text{const (blue line)} \text{ across expansion}$$

$$u_4 - \frac{2a_4}{\gamma-1} = u_p(t) - \frac{2a_p(t)}{\gamma-1} = u_3 - \frac{2a_3}{\gamma-1}$$

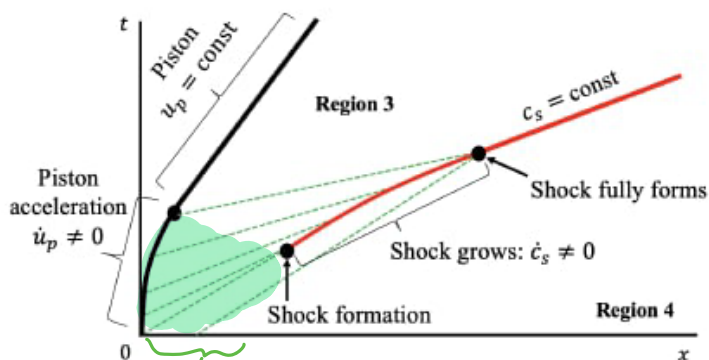
* function of time applies across blue highlighted region.

since $a_3 \geq 0$ to be physically permissible,

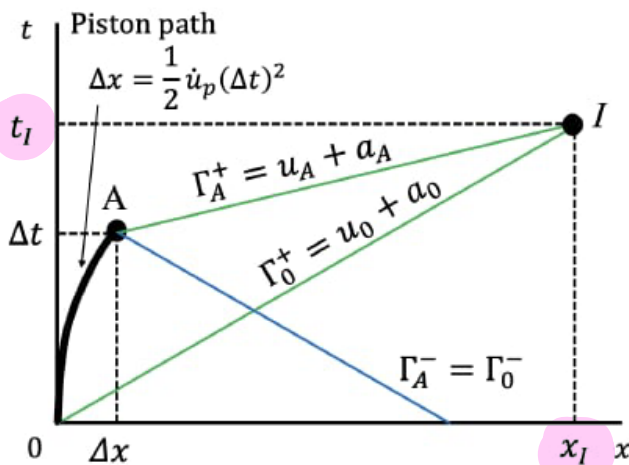
$$u_{3\min} = -\frac{2a_4}{\gamma-1}$$

this is the cavitation limit.

COMPRESSIONS



region up until shock forms still homentropic! so characteristics $P^\pm$ still hold



* Page 88 of course reader does a deeper dive into how to solve for t_I & x_I of initial shock formation using geometry.

Chapter 7: Burger's & Viscous Burger's Eqns.

inviscid Burger's: $\frac{\partial v}{\partial t} + v \frac{\partial v}{\partial x} = 0$

OR

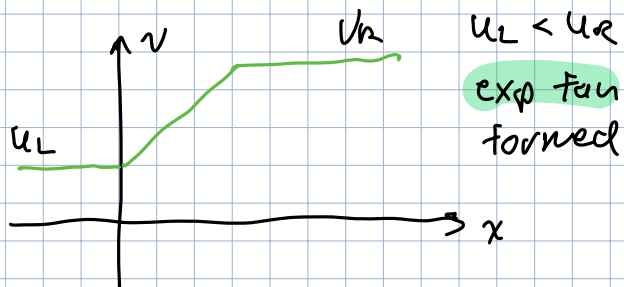
$$\frac{\partial v}{\partial t} + \frac{\partial}{\partial x} \left(\frac{1}{2} v^2 \right) = 0$$

* where $\gamma^\pm = \Gamma^\pm$ since $\gamma = 3$ & $-\frac{1}{\rho} \frac{\partial p}{\partial x} = 0$

x-v diagrams:



$U_L > U_R$,
shock will or
already formed



Add on questions to ↗

① when will shock form? t_f

x_L : @ $x_L = -1$, $t = 0$

x_R : @ $x_R = 1$, $t = 0$

→ when $x_L = x_R$

$$t = \frac{x+1}{U_L}$$

$$t = \frac{x-1}{U_R}$$

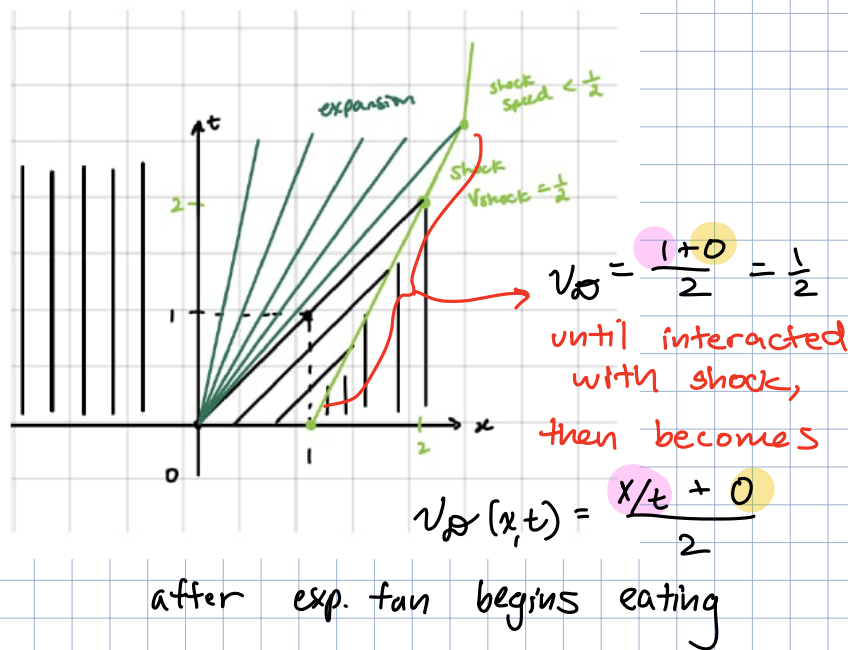
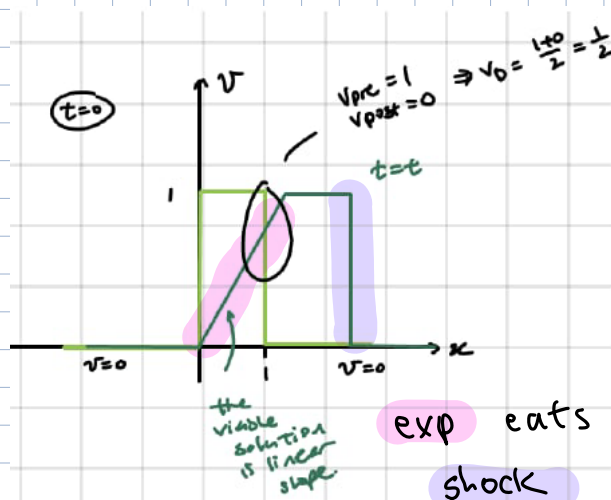
distance to ellipse!
 $t = \frac{2}{U_L - U_R}$

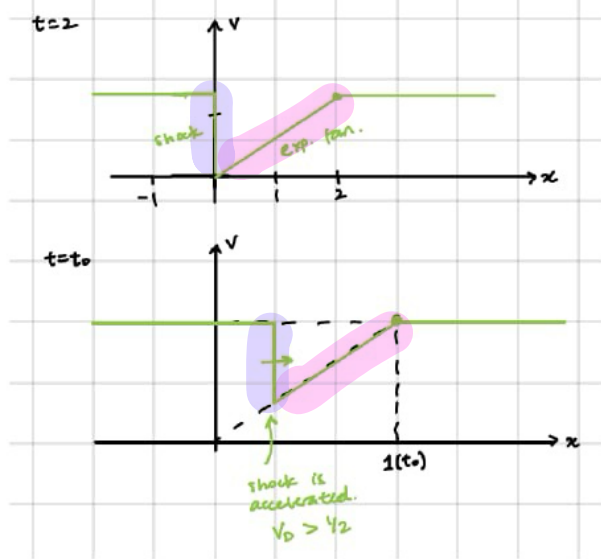
② where will shock form? x_f ? → self exercise

③ Speed of Resulting Shock?

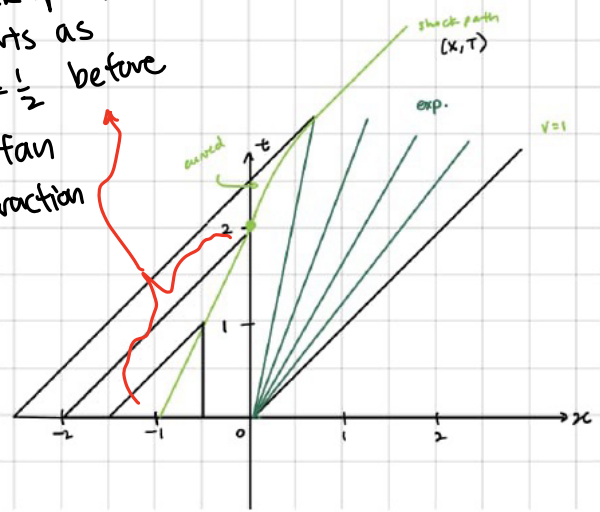
$$V_\sigma = \frac{\Delta F}{\Delta U} = \frac{\Delta \left[\frac{1}{2} v^2 \right]}{\Delta v} \Rightarrow V_\sigma = \frac{v_{\text{post}} + v_{\text{pre}}}{2}$$

Examples of $x=0$ centered exp fans EATING / GETTING EATEN





shock speed starts as $v_s = \frac{1}{2}$ before exp fan interaction



shock eats expansion

Burger's initial value problem

* see page 98 in book.

* only key result is

$$t_{\min} = - \frac{1}{\frac{dv}{dx_0} \big|_{x_0, \text{neg-max}}}$$

Convince yourself for $t > 2$,

$$v_s = \frac{1}{2} \left(1 + \frac{x}{t} \right)$$

hence: $\frac{dx}{dt} = v_s = \frac{1}{2} \left(1 + \frac{x}{t} \right)$

$\Rightarrow$ use @ $t=2, x=0$, as BC
@ $t \rightarrow \infty, v_s \rightarrow 1$

$X = \alpha \sqrt{t} + \beta t$ solve for α, β .

Viscous Burger's Eqn.

Shock-frame vars:

$w = v - v_s \rightarrow$ shock frame velocity

$z = x - v_s t \rightarrow$ shock frame position

Viscous Burgers becomes $\frac{\partial v}{\partial t} + v \frac{\partial v}{\partial x} = \frac{\partial}{\partial x} \left(\mu \frac{\partial v}{\partial x} \right) \Rightarrow w \frac{\partial w}{\partial z} = \mu \frac{\partial^2 w}{\partial z^2}$

let $w_0 = w(z=0)$, integrating PDE

$w(z) = - \sqrt{2\mu |w_0'|} \tanh \left(\sqrt{\frac{|w_0'|}{2\mu}} z \right)$ for $w_0' < 0$ (permissible)

$w(z) = \sqrt{2\mu |w_0'|} \tan \left(\sqrt{\frac{|w_0'|}{2\mu}} z \right)$ for $w_0' > 0$ (not permissible)

why? as $z \rightarrow \text{asympt.}$, $w \rightarrow \infty!$

$$w(z) = - \frac{w_L - w_R}{2} \tanh \left(\frac{w_L - w_R}{4\mu} z \right)$$

where $w_L = \lim_{z \rightarrow -\infty} w(z)$

$w_R = \lim_{z \rightarrow \infty} w(z)$

* equation of VISCOUS Burger's under shock-frame variables w, z .

Chapter 10 2D Supersonic flow (finally)

conservative matrix form

$$\text{start: } \frac{\partial U}{\partial t} + \frac{\partial F(U)}{\partial x} + \frac{\partial G(U)}{\partial y} = 0$$

2D Euler in cons. form.

where

$$U = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho E \end{bmatrix} \quad F = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho u v \\ \rho u H \end{bmatrix} \quad G = \begin{bmatrix} \rho v \\ \rho v u \\ \rho v^2 + p \\ \rho v H \end{bmatrix}$$

$\underbrace{\rho u}_{u \text{ advection}} \quad \underbrace{\rho v}_{v \text{ advection}}$

$$F = \begin{bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \end{bmatrix} = \begin{bmatrix} \frac{u_2}{u_1} \\ \frac{u_2^2}{u_1} + (\gamma - 1) \left(u_4 - \frac{1}{2} \frac{u_2^2}{u_1} - \frac{1}{2} \frac{u_3^2}{u_1} \right) \\ \frac{u_2 u_3}{u_1} \\ \frac{u_2}{u_1} \left\{ \gamma u_4 - (\gamma - 1) \frac{u_2^2}{2u_1} - (\gamma - 1) \frac{u_3^2}{2u_1} \right\} \end{bmatrix} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \text{explicit definition of } F(U) \rightarrow \text{use to find } A_c = \frac{\partial F}{\partial U} \text{ Jacobian}$$

$$G = \begin{bmatrix} G_1 \\ G_2 \\ G_3 \\ G_4 \end{bmatrix} = \begin{bmatrix} \frac{u_3}{u_1} \\ \frac{u_3^2}{u_1} + (\gamma - 1) \left(u_4 - \frac{1}{2} \frac{u_2^2}{u_1} - \frac{1}{2} \frac{u_3^2}{u_1} \right) \\ \frac{u_2 u_3}{u_1} \\ \frac{u_3}{u_1} \left\{ \gamma u_4 - (\gamma - 1) \frac{u_2^2}{2u_1} - (\gamma - 1) \frac{u_3^2}{2u_1} \right\} \end{bmatrix} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \text{explicit definition of } G(U) \rightarrow \text{use to find } B_c = \frac{\partial G}{\partial U}$$

end: $\Rightarrow \frac{\partial U}{\partial t} + A_c \frac{\partial U}{\partial x} + B_c \frac{\partial U}{\partial y} = 0$

primitive matrix form (*page 131 for full derivation)

$$\frac{\partial W}{\partial t} + A_w \frac{\partial W}{\partial x} + B_w \frac{\partial W}{\partial y} = 0 \quad \Rightarrow \quad \underbrace{\begin{pmatrix} \rho \\ u \\ v \\ p \end{pmatrix}_t}_W + \underbrace{\begin{pmatrix} u & \rho & 0 & 0 \\ 0 & u & 0 & 1/\rho \\ 0 & 0 & u & 0 \\ 0 & \rho a^2 & 0 & u \end{pmatrix}}_{A_w} \underbrace{\begin{pmatrix} \rho \\ u \\ v \\ p \end{pmatrix}_x}_x + \underbrace{\begin{pmatrix} v & 0 & \rho & 0 \\ 0 & v & 0 & 0 \\ 0 & 0 & v & 0 \\ 0 & 0 & \rho a^2 & v \end{pmatrix}}_{B_w} \underbrace{\begin{pmatrix} \rho \\ u \\ v \\ p \end{pmatrix}_y}_y = 0. \quad (10.3)$$

Chapter 11 - 2D inviscid Flow, Crocco's Theorem, vorticity

Crocco's Theorem:
$$\frac{\partial \vec{u}}{\partial t} + \left[\nabla \left(\frac{1}{2} |\vec{u}|^2 \right) - \vec{u} \times \vec{\omega} \right] + (\nabla h - T \nabla s) = 0$$

 OR
$$\frac{\partial \vec{u}}{\partial t} + \nabla H - T \nabla s = \vec{u} \times \vec{\omega}$$

where $\vec{\omega} := \nabla \times \vec{u}$ (vorticity)

Takeaway: flow is irrotational ($\vec{\omega} = \vec{0}$) if

- ① $\frac{\partial}{\partial t} = 0$ (steady)
 - ② $\nabla H = 0$ (isenthalpic)
 - ③ $\nabla s = 0$ (isentropic)
- } + ④ inviscid

Key idea: a velocity potential DNE for rotational flows.

If $\vec{\omega} = 0$, ϕ can be defined where $\vec{u} = \nabla \phi$ & $\nabla \times \nabla \phi = \vec{\omega} = 0$
 (curl of div = 0)

Steady - State irrotational flow

$$\begin{bmatrix} u \\ v \\ w \end{bmatrix} \cdot \left[\left(u \frac{\partial}{\partial x} + v \frac{\partial}{\partial y} + w \frac{\partial}{\partial z} \right) \begin{bmatrix} u \\ v \\ w \end{bmatrix} \right] = a^2 \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right) \quad (\star)$$

Linear Perturbations analysis.

let $u = u_0 + u'$ $v = v'$ $w = w'$ (where ' denotes a perturbation)
 on 1st-order $\sim O(\epsilon)$ where $\epsilon \ll 1$
 * important to understand that in perturbation analysis, order $\neq$ Taylor expansion order.

Substituting into $(\star)$ yields

$$(u_0^2 - a^2) \frac{\partial u'}{\partial x} - a^2 \frac{\partial v'}{\partial y} - a^2 \frac{\partial w'}{\partial z} = 2u_0 u' \frac{\partial u'}{\partial x} + O(\epsilon^2) + O(\epsilon^3) + \dots$$

div by a^2 both sides & drop sneaky

$$(M_\infty^2 - 1) \frac{\partial u'}{\partial x} - \frac{\partial v'}{\partial y} - \frac{\partial w'}{\partial z} = 0$$

all 1st-order perturbations

a sneaky 2nd-order retained

elliptic PDE

hyperbolic PDE

For $M_\infty \leq 0.1 \rightarrow \frac{\partial u'}{\partial x} + \frac{\partial v'}{\partial y} + \frac{\partial w'}{\partial z} = 0 \rightarrow \phi_{xx} + \phi_{yy} + \phi_{zz} = 0 \rightarrow \nabla^2 \phi = 0$

For $0.1 \leq M_\infty \leq 0.7 \rightarrow \phi_{\tilde{x}\tilde{x}} + \phi_{yy} + \phi_{zz} = 0$ where $\tilde{x} = \sqrt{1 - M_\infty^2} x$

For $0.7 \leq M_\infty \leq 1.4 \rightarrow (1 - M_\infty^2 - 2M_\infty \phi_x) \frac{\partial u'}{\partial x} + \frac{\partial v'}{\partial y} + \frac{\partial w'}{\partial z} = 0$

For $1.4 \leq M_\infty \leq 5 \rightarrow (1 - M_\infty^2) \frac{\partial u'}{\partial x} + \frac{\partial v'}{\partial y} + \frac{\partial w'}{\partial z} = 0$

For $M_\infty \geq 5 \sim O(\epsilon^2) \& O(\epsilon^3)$ come back into play $\rightarrow$ nasty hyperbolic